MODULE-III
ENGINEERING MATERIALS

Introduction, Classification, Ferrous (IRON) and Non-Ferrous metals (COPPER)
Types, Properties and their applications.

Introduction to Engineering Materials

- Engineering materials are substances used to create structures, machines, devices, and components.
- Selection depends on properties such as strength, hardness, ductility, conductivity, and cost.

Classification of Engineering Materials

- Metals (Ferrous & Non-Ferrous)
- Ceramics (Oxides, Carbides, Nitrides)
- Composites (Fiber-reinforced, Particle-reinforced)
- Advanced materials (Smart materials, Nano-materials)

Ferrous Metals

- Ferrous Metals –
  All ferrous metals are having iron as common element. All ferrous materials are having very high permeability which makes these materials suitable for construction of machines.
- Almost 90% of manufactured metals are ferrous metals.
- The word ‘ferrous’ comes from the Latin word ‘ferrum,’ which means ‘iron.’
- Ferrous metals include steel, cast iron, as well as alloys of iron with other metals (such as stainless steel)

Types of Ferrous Metals

- Cast Iron – brittle, good compressive strength.
- Mild Steel – low carbon, ductile, weldable.
- High Carbon Steel – harder and stronger.
- Alloy Steel – improved mechanical properties by alloying.

Ferrous Metals : IRON

- Iron is the most commonly used and cheapest metal. There are three types: pig iron, cast iron and wrought iron.
- Pig iron : a crude form of iron, used as a raw material for the production of various other ferrous metals, such as cast iron and steel.
- Cast iron: created by melting pig iron with coke and limestone. Cast iron tends to be brittle and is notoriously difficult to weld. However, it is an ideal engineering material with a wide range of applications particularly in the automotive industry. This is due to its relatively low melting point, castability, excellent machinability and resistance to both deformation and wear.
- Wrought iron: tough, ductile and corrosion-resistant, wrought iron is a type of highly purified metal, with small amounts of silicate slag moulded into the filaments. This means it is ideal for forging into products such as guard rails, gates and garden furniture.

Properties of Ferrous Metals

- Good Ductility and Toughness
- High Tensile Strength
- High Thermal and Electrical Conductivity
- Low corrosion resistance
- Recyclable
- Magnetic Properties

Applications of Ferrous Metals

- Construction (beams, rebar)
- Machine components (shafts, gears)
- Automobiles (engine blocks, chassis)
- Tools and cutting instruments

Non-Ferrous Metals

- Non-Ferrous Metals –
  All non-ferrous metals are having very low permeability.
- Non-ferrous metals are alloys or metals containing zero (or very little) iron. They tend to be more expensive than ferrous metals due to their lighter weight.
- Example: Silver, Copper, Gold, Aluminum etc.

Types of Non-Ferrous Metals

- Aluminum – lightweight, corrosion resistant
- Copper – highly conductive
- Zinc – used in galvanizing
- Lead – soft, corrosion resistant
- Titanium – high strength-to-weight ratio

Non-Ferrous Metal : Copper

- Copper is a soft, malleable and ductile metal with a reddish-brown appearance. It provides good conductivity of electricity, can be cast, forged, rolled and drawn into wires. It is used extensively for wire, coins and household utensils.

Properties of Non-Ferrous Metals

- Non-magnetic properties
- Easy to fabricate
- High resistance to corrosion
- Good thermal and electrical conductivity
- Low density

Applications of Non-Ferrous Metals

- Electrical wiring (copper)
- Aerospace structures (titanium, aluminum)
- Batteries (lead)
- Galvanizing steel (zinc)
- Packaging (aluminum foils, cans)

COMPOSITE MATERIALS

Introduction,
Constituents of a composite
Classification of Matrix
Types of Matrix
And Reinforcement materials,
Advantages, Disadvantages and
Applications of composite materials.

Introduction

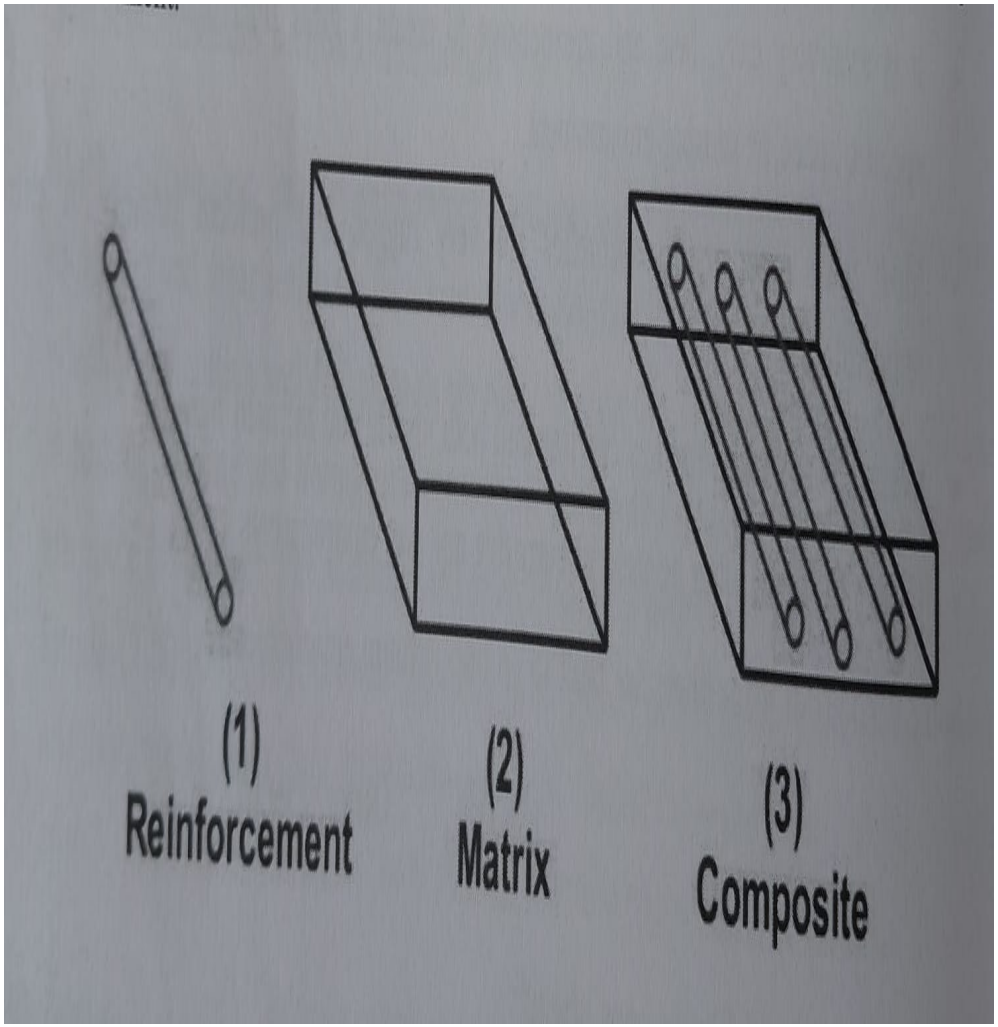
- Composite materials is a combination of two or more dissimilar materials with significantly different physical or chemical properties between them and the resulting material properties are enhanced compared to the individual constituent materials.
- In composites one of the materials, called the reinforcing phase, is in the form of fibers, sheets, or particles, and is embedded in the other materials called the matrix phase.
- The reinforcing material and the matrix material can be metal, ceramic, or polymer

Constituents of a composite

- The composite material comprises two main components.
- In a composite material one material forms a matrix to bond together the other material called reinforcement material.

1) The reinforcements, the continuous or discontinues phase used to strengthen the composite.
2) The matrix is the continuous surrounding material in which the reinforcement is embedded.

- Example :- Glass fiber reinforcement

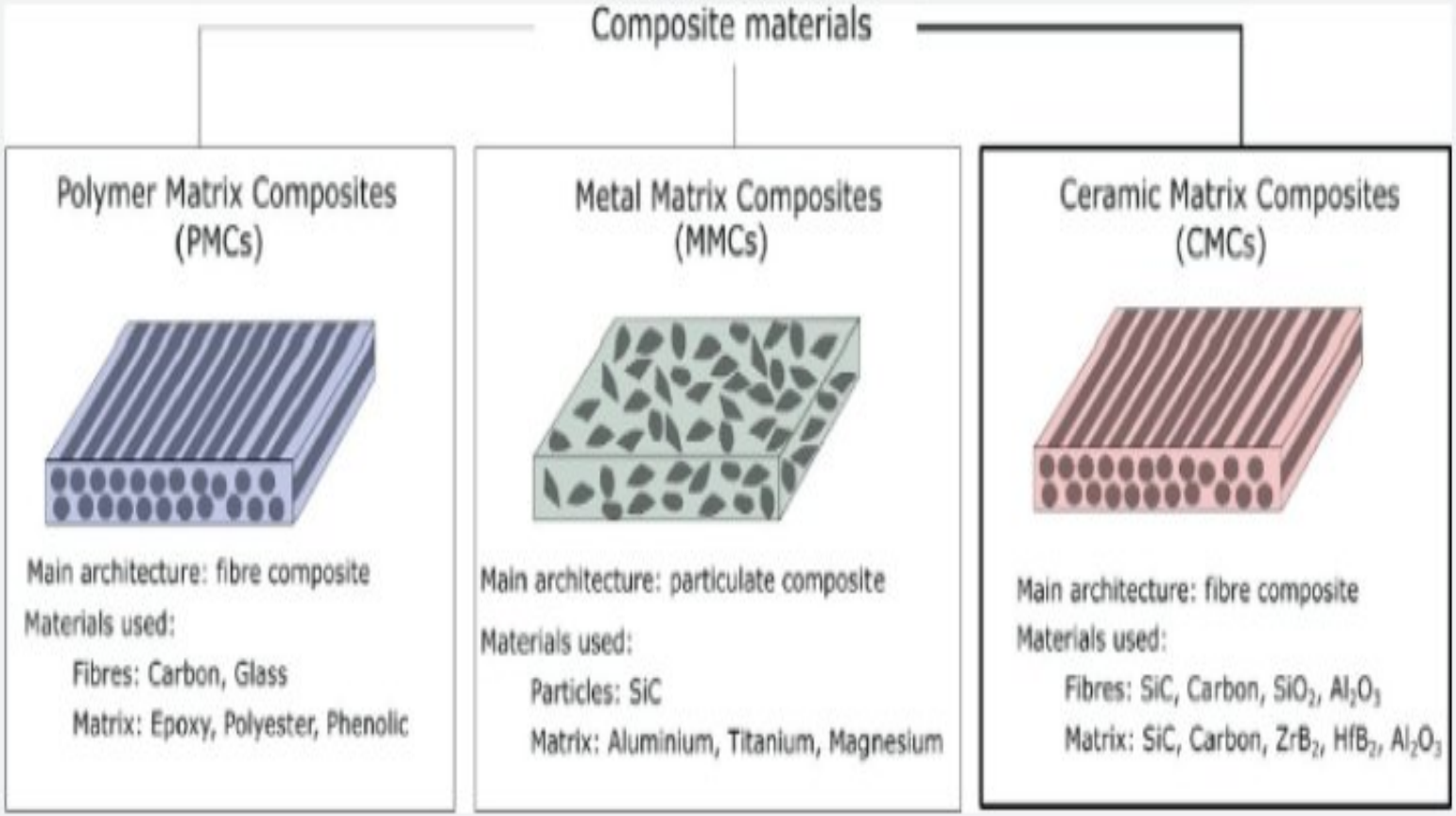


Classification of Composites

- The classification of composite materials is based on the composition and structure of the components involved.

1) Composite Materials Based on Matrix Type:
a) Polymer Matrix Composites (PMC)
b) Metal Matrix Composites (MMC)
c) Ceramic Matrix Composites (CMC)
2) Composite Materials Based on Reinforcement Type:
a) Fiber-Reinforced Composites (FRC)
b) Particulate Reinforced Composites
c) Structural or laminate Composites

Composite Materials Based on Matrix Type



Composite Materials Based on Matrix Type

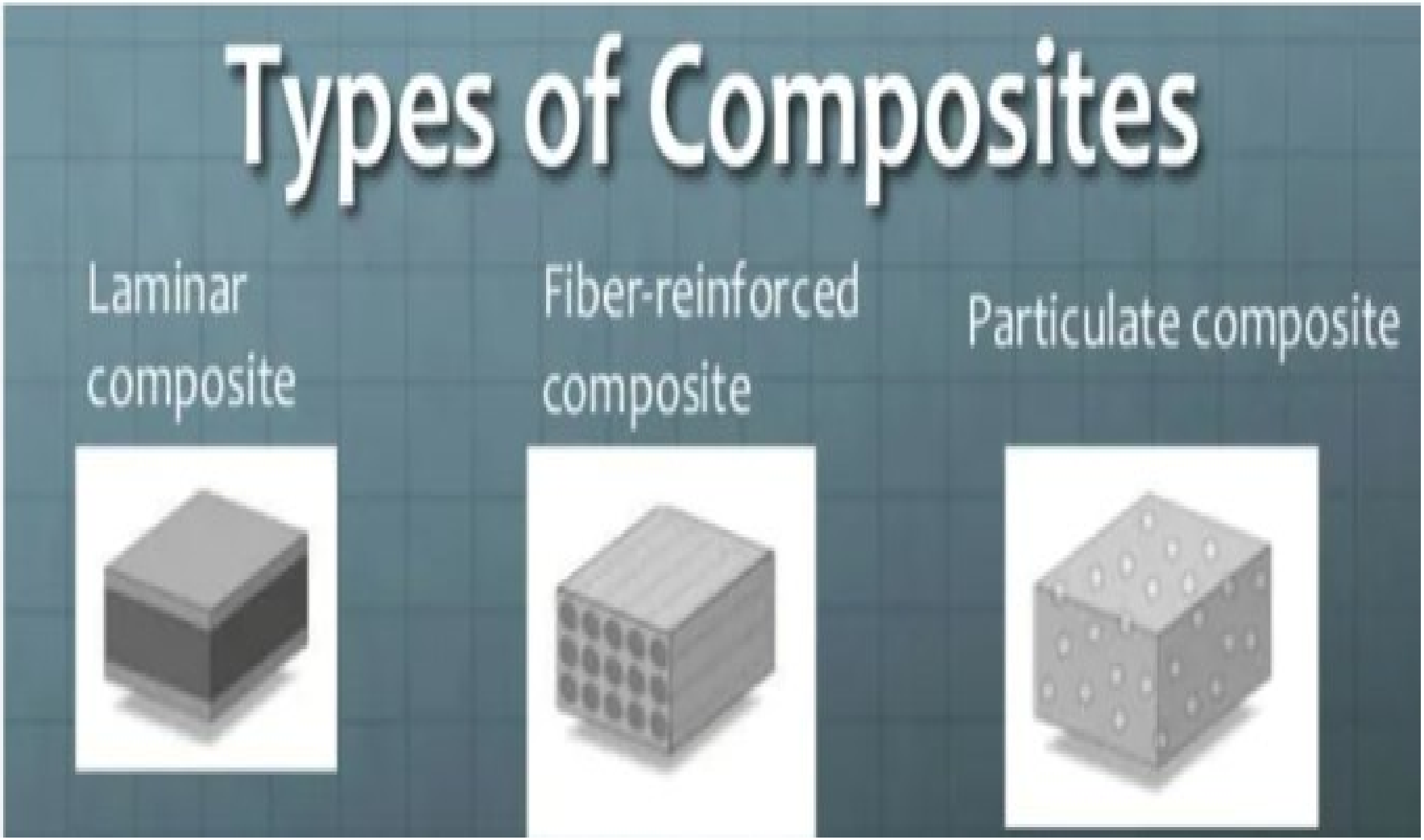
- The matrix material plays a critical role in binding the reinforcement together and protecting it from environmental damage.

a) Polymer Matrix Composites (PMC) are the most widely used composite materials, consisting of a polymer resin as the matrix combined with fibers like glass and carbon. PMCs are preferred for industries needing lightweight yet strong materials, such as the automotive and sports equipment sectors.

b) Metal Matrix Composites (MMC) use metals like aluminum, magnesium, or titanium as the matrix material. Reinforced with ceramic or metal fibers, they offer enhanced thermal resistance and strength, making them ideal for aerospace and automotive applications.

c) Ceramic Matrix Composites (CMC) consist of ceramic materials reinforced with ceramic fibers. They offer exceptional heat and corrosion resistance, making them ideal for extreme environments like aerospace and defense sectors.

Composite Materials Based on Reinforcement Type



Composite Materials Based on Reinforcement Type

- Reinforcement provides additional strength and stiffness to the matrix material.

a) Fiber-reinforced composites (FRC) consist of strong fibers, such as glass, carbon embedded in a matrix material like polymer, metal, or ceramic. These fibers act as the primary load carriers, while the matrix binds them together, ensuring stress distribution and protection from environmental factors.

b) Particulate reinforced composites are materials where small particles, such as ceramics, metal oxides, or carbides, are uniformly dispersed within a matrix material like polymer, metal, or ceramic. These particles help enhance properties such as hardness, wear resistance, and compressive strength.

c) Structural or laminate composites are advanced materials that combine multiple layers of composites or core materials to create strong, load-bearing structures. They are typically designed with a core material, such as foam or honeycomb, sandwiched between two thin, stiff composite face sheets. This layered structure enhances mechanical properties by distributing loads evenly across the material while maintaining a low weight.

Advantages, Disadvantages of composite materials

ADVANTAGES

1) Composites are light and reduce overall weight.
2) They offer excellent strength-to-weight ratios.
3) They resist corrosion and wear.
4) Easy to shape and customize.
5) Provide electrical and thermal insulation.
6) Resist chemical damage.
7) Absorb and withstand impacts.

DISADVANTAGES

1) Manufacturing composites can be expensive.
2) Repairs may require specialized techniques.
3) Layers can separate under stress.
4) Some composites are not fire-resistant.
5) Recycling composites can be difficult.

Applications of composite materials

1) Durable in building structures.
2) Used in Sports: tennis rackets, helmets etc.
3) Resistant boat hulls and parts.
4) Makes robust turbine blades.
5) Corrosion-resistant pipelines.
6) Reinforce bridges and buildings.
7) Solar panel components.
8) Aerospace: Aircraft wings, panels

SMART MATERIALS

Introduction,
Types - Piezoelectric materials,
Advantages, Disadvantages
and Applications

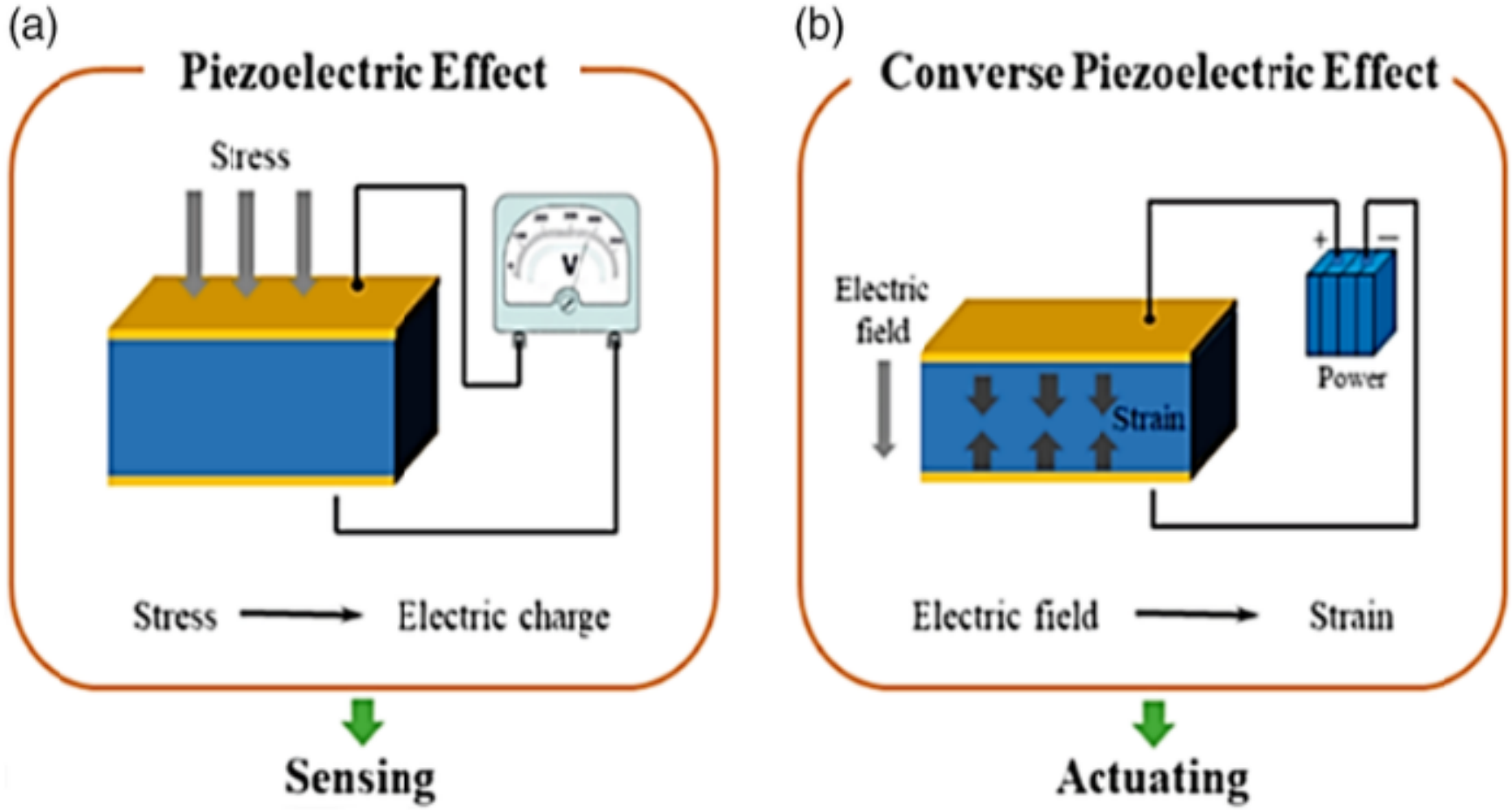
Introduction

- Smart materials are materials that can sense changes in their environment (such as temperature, pressure, electric field, magnetic field, stress, etc.) and respond to those changes in a controlled and reversible manner.
- These materials consists of inherent intelligence to adapt to external stimuli.
- They have one or more property changes that occur when an external stimulus is applied. These responses may include changes in shape, stiffness, color, viscosity, or electrical properties.

Types of Smart Materials

- Smart materials can be divided into type 1 and type 2 smart material.
- Type 1 :- Materials exhibit change in the properties like chemical, magnetic, mechanical, electrical with response to change in the external stimuli. The property change occurs as a result of internal energy input which alters material microstructure.
- Examples :- phototropic Materials, Electrochromic Materials, Thermotropic Materials
- type 2 :- Materials exhibit change in the properties with response to change in the external stimuli. The property changes occurs as a result of changes in the energy state of the material composition, but material remains the same.
- Example :- Piezoelectric Materials, Shape Memory Alloys , Light Emitting Materials

Piezoelectric materials



Piezoelectric materials

- The Piezo word derived from Greek word which means electricity resulting from pressure.
- Piezoelectric materials are a class of smart materials that generate an electric charge when mechanical stress is applied. This property is known as the piezoelectric effect.
- Important property of the Piezoelectric material is that it displays both direct Piezoelectric effect.
- Direct Piezoelectric effect :- The generation of electricity when an stress is applied.
- Converse Piezoelectric effect :- The generation of stress when an electric field is applied.

Advantages, Disadvantages Smart Materials

Advantages

1) Improved Efficiency and Adaptability
2) Reduced Maintenance Costs
3) Enhanced Safety and Durability
4) Compact and Lightweight Design

Disadvantages

1) Initial cost much higher than traditional materials.
2) Manufacturing and Scalability Issues
3) Limitations in extreme conditions
4) Complexity

Applications of Smart Materials

1) Aerospace Applications :- Vibration control in aircraft structures using piezoelectric actuators.
2) Automotive Applications :- Smart braking systems with pressure-sensitive sensors.
3) Medical & Healthcare Application :- Ultrasound imaging transducers using piezoelectric materials
4) Civil Engineering & Construction
5) Structural health monitoring of bridges, dams, and buildings
6) Defense & Military Applications :- Acoustic sensors for submarine communications.
7) Consumer Products :- Photochromic Materials Found in light-sensitive eyeglasses that darken automatically when exposed to UV light outside.